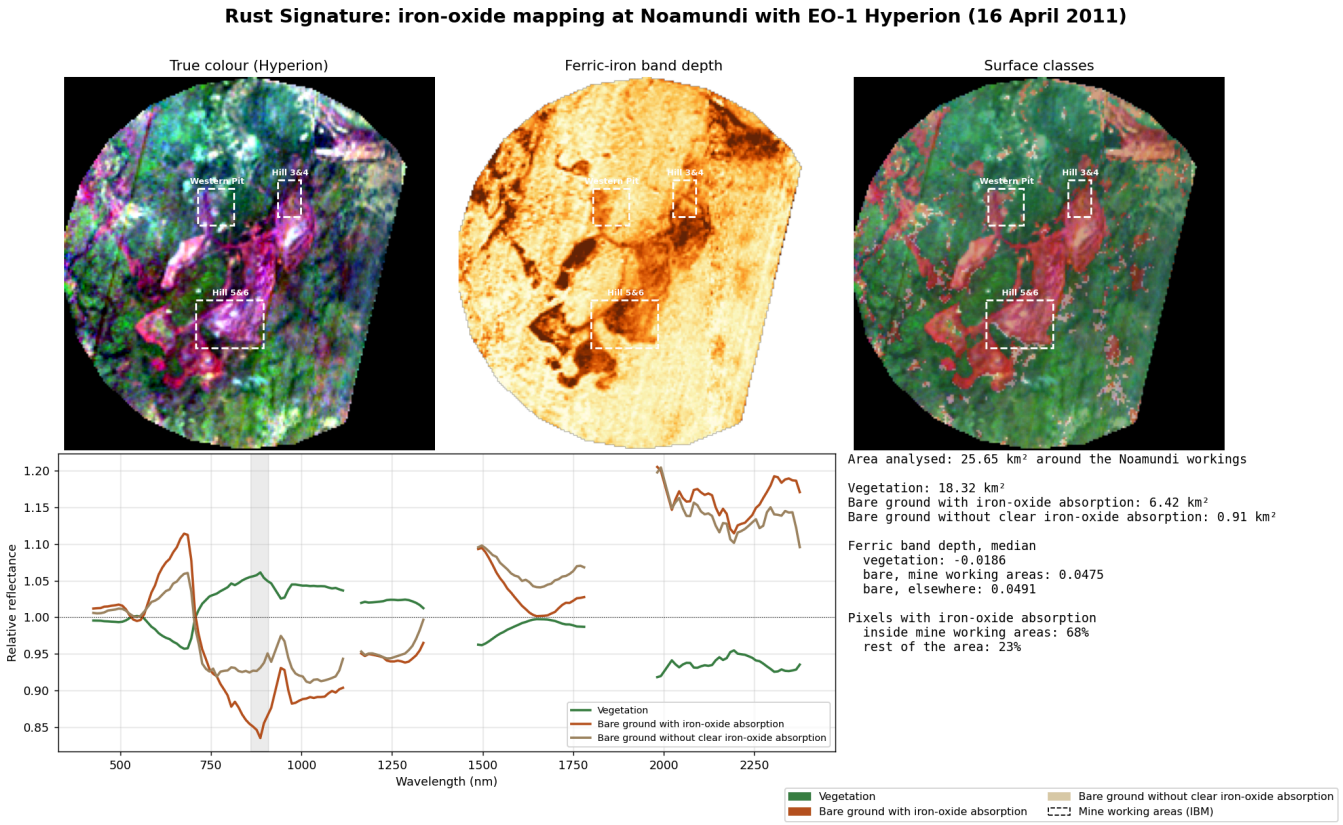
Rust Signature: iron-oxide mapping at Noamundi with EO-1 Hyperion

A short case study. I take one EO-1 Hyperion scene (16 April 2011) over the Noamundi iron-ore mine in Jharkhand, India, and ask a simple question: does the ferric-iron absorption near 900 nm show up where the mine is?



What I did

1. **Data.** Hyperion scene of 16 April 2011 from Google Earth Engine (`EO1/HYPERION`), clipped to a 3 km radius around the workings (`scripts/00_gee_export.js`). The scene edge cuts off the eastern part of the circle, so 25.65 km² is analysed.
2. **Radiance.** Earth Engine stores Hyperion as scaled integers, so I divide by 40 (VNIR, B008–B057) and by 80 (SWIR, B077–B224) to get radiance.
3. **Band clean-up.** I drop the water-vapour and CO_2 bands (1115–1145, 1340–1470, 1790–1970, 1995–2020 nm), the SWIR bands that repeat the end of the VNIR range, and bands that are empty or drop out. 157 of 198 bands remain. Vertical detector stripes are evened out by matching each column's mean and spread to the band's (moment matching).
4. **Relative reflectance.** I use IARR (Kruse, 1988): each pixel's radiance divided by the scene-mean radiance in that band. It is simple and needs no atmospheric model, but the result is relative to this scene.
5. **Indices.**
    - NDVI from red (640–680 nm) and NIR (820–880 nm) radiance.

    - Ferric-iron band depth: 1 − centre / continuum. The centre is 860–910 nm and the straight-line continuum runs between 740–780 nm and 990–1030 nm.
6. **Classes.**
    - Vegetation: NDVI above an Otsu threshold.
    - Bare ground with iron-oxide absorption: ferric band depth > 0, i.e. a dip deeper than the scene average.
    - Bare ground without clear absorption: the rest.
7. **Check against the mine.** I compare the classes with the three working areas listed by corner coordinates in the Indian Bureau of Mines inspection report for the mine: Hill 3&4, Hill 5&6 and the Western Pit.

I run the whole analysis with one command, `python run_all.py`, on the exported tif in this folder.

Results

Area analysed	25.65 km²
Vegetation	18.32 km²
Bare ground with iron-oxide absorption	6.42 km²
Bare ground without clear iron-oxide absorption	0.91 km²
Pixels with iron-oxide absorption, inside the mine working areas	68%
Pixels with iron-oxide absorption, rest of the area	23%
Median ferric band depth: vegetation / bare in working areas / bare elsewhere	−0.019 / 0.048 / 0.049

- The bare ground shows a clear absorption trough centred near 880–900 nm, the shape expected for ferric iron (hematite, goethite). Vegetation shows none.
- The mine working areas are about three times as likely as the rest of the area to fall in the iron-oxide class.
- Inside bare ground, the working areas look the same as the other bare ground (median depth 0.048 vs 0.049). The pits, benches and waste dumps around them all carry the iron signal, so at 30 m the three listed pits cannot be told apart from the rest of the mining complex.

Limits

- One date (2011), 30 m pixels, no field or lab spectra. The class is "bare ground with an iron-oxide absorption", not an ore-grade map.
- IARR is relative to the scene mean, and this scene is mostly forest. So "deeper than average" partly means "less vegetated". Ferric depth and NDVI correlate at r = −0.80.
- The working areas are rectangles built from corner coordinates, so their edges are approximate.
- Hyperion's SWIR signal is noisy, so I do not use the 2200 nm clay feature for any class.

Files

- `scripts/00_gee_export.js`: Earth Engine export
- `scripts/01_prepare.py`: radiance, band clean-up, destriping, IARR, band QA plot

- `scripts/02_indices_and_classes.py`: NDVI, ferric depth, classes, working-area check, `outputs/summary.json`
- `scripts/03_figures_and_map.py`: maps, poster, interactive map (`outputs/noamundi_classification_map.html`) and plots

References

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